Arithmetic autocorrelation of binary half-ℓ-sequences with connection integer $p^r q^s$

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Abstract

Half-ℓ-sequences, as a extension of ℓ-sequences, have attracted research interest over the past decade. The arithmetic correlation of half-ℓ-sequences is known for connection integers of the form p^r. In this paper, we extend this result by deriving the arithmetic correlation for half-ℓ-sequences with connection integers of the form $p^r q^s$. The results indicate that when $p \equiv -1 \pmod 8$ and $q \equiv \pm 3 \pmod 8$, the arithmetic autocorrelation can be determined by the number of odd integers in the cyclic subgroup generated by 2 modulo p.

Keywords: binary sequence, half-ℓ-sequence, arithmetic correlation, feedback with carry shift register

1 Introduction

Pseudorandom sequences are of paramount importance in modern communication and information security systems. Among these, Linear Feedback Shift Register (LFSR) sequences have been the subject of extensive research due to their capability for high-speed sequence generation enabled by efficient hardware implementation. In the 1990s, Goresky and Klapper introduced the feedback with carry shift register (FCSR) sequences, which exhibit an algebraic theory analogous to that of LFSRs [1,2].

Let $\mathcal{S} = \{s_i\}_{i=0}^{\infty}$ be a binary sequence of period T and every $s_i \in \{0,1\}$. Then $\mathcal{S}$ uniquely corresponds to a rational number

$$\alpha = \sum_{i=0}^{\infty} = -\frac{\sum_{i=0}^{T-1} s_i 2^i}{2^T - 1} = -\frac{m}{n},$$

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where $0 < m < n$ and m is coprime to n. And we called n is a connection integer of $\mathcal{S}$. And $\mathcal{S}$ can be defined equivalently by $s_i = \left(f2^{-i} \bmod n\right) \bmod 2$. Thus $T = \mathrm{ord}_n(2)$, is the order of 2 modulo n [2].

The τ-shift of $\mathcal{S}$, denoted by $\mathcal{S}^{(\tau)}$, corresponds to a rational number, denotes $\mathcal{S}^{(\tau)} = -\frac{m_\tau}{n}$, with $m_\tau = 2^{T-\tau}m \pmod n$, $0 < m_\tau < n$ and m_τ is coprime to n [2]. Hence

$$\alpha - \alpha_{(\tau)} = -\frac{m}{n} + \frac{m_\tau}{n} = c + \left(-\frac{\left(2^{T-\tau} - 1\right) m \bmod n}{n}\right) = c - \frac{m'}{n'},$$

for some integer c, where $0 < m' < n'$ and m' is coprime to n'. Consequently, the sequence generated by $\alpha - \alpha_{(\tau)}$ (denoted $\mathcal{U}$) shares an identical periodic component with the sequence associated with $\frac{m'}{n'}$.

As a main measure for FCSRs, the arithmetic correlation is defined as the carry-based counterpart to the classical notion of correlation. For detail, the arithmetic auto-correlation of $\mathcal{S}$ at τ, denoted by $A^A_{\mathcal{S}}(\tau)$, defined by

$$A^A_{\mathcal{S}}(\tau) = T - 2\frac{T}{\mathrm{ord}_{n'}(2)} \cdot \lambda' = \frac{\mathrm{ord}_n(2)}{\mathrm{ord}_{n'}(2)}(\mathrm{ord}_{n'}(2) - 2\lambda'), \tag{1}$$

where λ' is the number of ones in a period of $\mathcal{U}$, see [2–5].

ℓ-sequences are maximal-length FCSR sequences with connection integer n and period $T = \mathrm{ord}_n(2) = \varphi(n)$ (Euler's totient function). A key property is their ideal arithmetic correlation, meaning the correlation equals zero under any non-trivial cyclic shift [4]. With ongoing research, Gu and Klapper have investigated half-ℓ-sequences, which achieve a period of $T = \varphi(n)/2$ with connection integer n [6]. Extensive research has been conducted on the properties of half-ℓ sequences [6–12]. In particular, for an odd prime power $n = p^r$, Chen et al. [11] studied the arithmetic correlations of half-ℓ-sequences. More recently, Xiao et al. [12] established the arithmetic auto-correlations for the cases $n = 3^r \times 5^s$ and $n = 7^r \times q^s$, showing that the maximum values are $T/2$ and $T/3$, respectively (with T being the period).

Motivated by the aforementioned studies, it is of interest to determine the arithmetic auto-correlation of binary half-ℓ-sequences with more general connection integers of the form $n = p_1^{r_1} p_2^{r_2} \dots p_k^{r_k}$, where the p_i are distinct primes. However, it follows easily that $T = \varphi(n)/2$ implies the connection integer n must be of the form $n = p^r q^s$ with $r \geq 1$ and $s \geq 1$. And by [13, Thm. 4], when $p \equiv q \equiv 3 \pmod 8$ or $p \equiv q \equiv 5 \pmod 8$, the binary half-$\ell$-sequence with connection integer $n = p^r q^s$ has ideal arithmetic auto-correlation.

Therefore, this work is to consider the remaining possible cases for binary half-ℓ-sequences with connection $n = p^r q^s$ that satisfies

$$p \equiv -1 \pmod 8 \text{ and } q \equiv \pm 3 \pmod 8,$$

or

$$p \equiv 3 \pmod 8 \text{ and } q \equiv -3 \pmod 8,$$

and determine their arithmetic auto-correlation.

2 Half-ℓ-Sequences with Connection Integers $n = p^r \times q^s$

In the subsequent discussion, let $(\frac{\cdot}{\cdot})$ denotes the Legendre symbol, $\mathrm{ord}_n(2)$ denotes the order of 2 modulo n, $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$, and $\mathbb{Z}_n^*$ denote the multiplicative group of units modulo n.

2.1 When $p \equiv -1 \pmod 8$ and $q \equiv \pm 3 \pmod 8$

we suppose that $p \equiv -1 \pmod 8$ and and $q \equiv \pm 3 \pmod 8$ let the sequences with connection integer $n = p^r q^s$ are half-ℓ-sequences. It follows that it satisfies $\mathrm{ord}_{p^r}(2) = \varphi(p^r)/2$ and $\mathrm{ord}_{q^s}(2) = \varphi(q^s)$.

Let $\mu = e^{2\pi i/p}$. Since $p \equiv -1 \pmod 8$, we note $D_p = \{2^i \pmod p : i = 0, 1, \ldots, \mathrm{ord}_p(2)\}$ is the set of quadratic residues modulo p, and we use $\mathbb{Z}_p^* \backslash D_p$ denotes the set of quadratic non-residue. For $p \nmid t$, by means of the quadratic Gauss sum modulo prime p (see [8]), and the Legendre symbol is completely multiplicative, we get

$$\sum_{v \in D_p} \mu^{tv} = -\frac{1}{2} + \frac{i}{2}\left(\frac{t}{p}\right)\sqrt{p}, \qquad \sum_{v \in \mathbb{Z}_p^* \backslash D_p} \mu^{tv} = -\frac{1}{2} - \frac{i}{2}\left(\frac{t}{p}\right)\sqrt{p}. \tag{2}$$

Lemma 2.1 *Let p is a prime satisfying $p \equiv -1 \pmod 8$ and $\mathrm{ord}_p(2) = \varphi(p)/2$. Then*

$$\sum_{v \in \mathbb{Z}_p^* \backslash D_p} \frac{1}{\mu^v + 1} = \frac{p-1}{4} - \frac{p-1}{4}\sqrt{p}\, i + \sqrt{p}\, i\Delta,$$

$$\sum_{v \in D_p} \frac{1}{\mu^v + 1} = \frac{p-1}{4} + \frac{p-1}{4}\sqrt{p}\, i - \sqrt{p}\, i\Delta,$$

where $\mu = e^{2\pi i/p}$, Δ is the number of odds in D_p.

Proof. Using the identity

$$\frac{2}{1+\mu^v} = \frac{1-(-\mu^v)^p}{1+\mu^v} = \sum_{t=0}^{p-1}(-1)^t \mu^{tv},$$

then by (2), we have

$$\begin{aligned}
\sum_{v \in \mathbb{Z}_p^* \backslash D_p} \frac{1}{\mu^v + 1} &= \frac{1}{2} \sum_{v \in \mathbb{Z}_p^* \backslash D_p} \sum_{t=0}^{p-1} (-1)^t \mu^{tv} \\
&= \frac{1}{2}\left[\frac{p-1}{2} - \Delta(-\frac{1}{2} - \frac{i}{2}\sqrt{p}) + (\frac{p-1}{2} - \Delta)(-\frac{1}{2} - \frac{i}{2}\sqrt{p})\right. \\
&\quad \left. -(\frac{p-1}{2} - \Delta)(-\frac{1}{2} + \frac{i}{2}\sqrt{p}) + \Delta(-\frac{1}{2} + \frac{i}{2}\sqrt{p})\right] \\
&= \frac{p-1}{4} - \frac{p-1}{4}\sqrt{p}\, i + \sqrt{p}\, i\Delta,
\end{aligned}$$

and from $\frac{1}{\mu^v+1} + \frac{1}{\mu^{p-v}+1} = 1$, we obtain the result. $\square$

Lemma 2.2 *Let $n = pq$, where p and q are primes satisfying $p \equiv -1 \pmod 8$ and $q \equiv \pm 3 \pmod 8$, respectively, and $\mathrm{ord}_n(2) = \varphi(n)/2$. Then the number of odd integers in $D_n = \{2^i \pmod n) : i = 0, 1, \ldots, \mathrm{ord}_n(2)\}$, denoted by $o_{1,1}$, is*

$$o_{1,1} = \begin{cases} (p-1)(q-1)/4, & \text{if } \left(\frac{q}{p}\right) = 1; \\ (p-1)(q-1)/4 - (p-1)/2 + 2\Delta, & \text{if } \left(\frac{q}{p}\right) = -1, \end{cases}$$

where Δ is the number of odds in D_p.

Proof. Let $\eta = \lambda\mu$ with $\lambda = e^{2\pi i/q}$ and $\mu = e^{2\pi i/p}$. From the standard exponential sums (see [8]), the number of even numbers in D_{pq} is given by

$$\begin{aligned}(p-1)(q-1)/2 - o_{1,1} &= \sum_{u\in D_{pq}} \frac{1}{pq} \sum_{b=1}^{(pq-1)/2} \sum_{t=0}^{pq-1} \eta^{t(u-2b)} \\ &= \frac{(pq-1)(p-1)(q-1)}{4pq} + \frac{1}{pq} \sum_{u\in D_{pq}} \sum_{t=1}^{pq-1} \eta^{tu} \sum_{b=1}^{(pq-1)/2} \eta^{-2tb} \\ &= \frac{(pq-1)(p-1)(q-1)}{4pq} + \frac{1}{pq} \sum_{u\in D_{pq}} \sum_{t=1}^{pq-1} \left(-\frac{1}{\eta^t+1}\right) \eta^{tu} \\ &= \frac{(pq-1)(p-1)(q-1)}{4pq} - \frac{1}{pq} \sum_{t=1}^{pq-1} \frac{1}{\eta^t+1} \sum_{u\in D_{pq}} \eta^{tu}.\end{aligned}$$

To evaluate this formula, we analyze the set $\{1, 2, \ldots, pq-1\}$ by dividing it into 3 cases. **Case (i)**: Let $P_1 = \{q, 2q, \ldots, (p-1)q\}$. We have

$$\begin{aligned}\sum_{t\in P_1} \frac{1}{\eta^t+1} \sum_{u\in D_{pq}} \eta^{ut} &= (q-1) \sum_{y=1}^{p-1} \frac{1}{\mu^{yq}+1} \sum_{v\in D_p} \mu^{vyq} \\ &= -\frac{1}{2}(q-1) \sum_{x=1}^{p-1} \frac{1}{\mu^x+1} + \frac{i\sqrt{p}(q-1)}{2} \sum_{x=1}^{p-1} \left(\frac{x}{p}\right) \frac{1}{\mu^x+1} \\ &= -\frac{(p-1)(q-1)}{4} + \frac{i\sqrt{p}(p-1)}{2} \left(\sum_{x\in D_p} \frac{1}{\mu^x+1} - \sum_{x\in \mathbb{Z}_p^*\setminus D_p} \frac{1}{\mu^x+1} \right) \\ &= -\frac{(p-1)(q-1)}{4} + \frac{i\sqrt{p}(p-1)}{2} \left(\frac{p-1}{2} - 2 \sum_{x\in \mathbb{Z}_p^*\setminus D_p} \frac{1}{\mu^x+1} \right) \\ &= (q-1)(\frac{1-p^2}{4} + p\Delta),\end{aligned}$$

where we use $\frac{1}{\mu^x+1} + \frac{1}{\mu^{p-x}+1} = 1$ and Lemma 2.1.

Case (ii): Let $P_2 = \{p, 2p, \ldots, (q-1)p\}$. Since $\psi(D_{pq}) = D_p \times \mathbb{Z}_q^*$, where ψ: $\mathbb{Z}_{pq} \longrightarrow$

$\mathbb{Z}_p \times \mathbb{Z}_q$ is a isomorphism. Then $D_{pq} \bmod q = \{\omega : u \in D_{pq}\} = \mathbb{Z}_q^*$. Thus

$$\sum_{t\in P_2} \frac{1}{\eta^t+1} \sum_{u\in D_{pq}} \eta^{ut} = \sum_{y=1}^{q-1} \frac{1}{\lambda^{py}+1} \cdot \frac{p-1}{2} \sum_{\omega\in\mathbb{Z}_q^*} \lambda^{\omega py} = -\frac{p-1}{2} \sum_{y=1}^{q-1} \frac{1}{\lambda^{py}+1} = -\frac{(p-1)(q-1)}{4},$$

where the last step follows from $\frac{1}{\lambda^{py}+1} + \frac{1}{\lambda^{q-py}+1} = 1$.

Case (iii): For $t \in \mathbb{Z}_{pq}^*$, since $\psi(D_{pq}) = D_p \times \mathbb{Z}_q^*$, it follows from (2) that,

$$\sum_{u\in D_{pq}} \eta^{ut} = \sum_{v\in D_p} \mu^{tv} \sum_{\omega\in\mathbb{Z}_q^*} \lambda^{t\omega} = -\sum_{v\in D_p} \mu^{tv} = \frac{1}{2} - \frac{i}{2}\left(\frac{t}{p}\right)\sqrt{p}.$$

Since $pq - t \equiv -t(\bmod pq)$, and -1 is a quadratic non-residue modulo p, it follows that $t \in D_{pq}$ iff $pq - t \in \mathbb{Z}_{pq}^* \backslash D_{pq}$. And note $t \in D_{pq}$ iff $t \pmod p \in D_p$, that is $\left(\frac{t}{p}\right) = 1$, and $t \in \mathbb{Z}_{pq}^* \backslash D_{pq}$ iff $t \pmod p \in \mathbb{Z}_p^* \backslash D_p$, that is $\left(\frac{t}{p}\right) = -1$. Thus by (2),

$$\begin{aligned}
\sum_{t\in\mathbb{Z}_{pq}^*} \frac{1}{\eta^t+1} \sum_{u\in D_{pq}} \eta^{ut} &= \sum_{t\in D_{pq}} \frac{1}{\eta^t+1} \sum_{u\in D_{pq}} \eta^{ut} + \sum_{t\in\mathbb{Z}_{pq}^*\backslash D_{pq}} \frac{1}{\eta^t+1} \sum_{u\in D_{pq}} \eta^{ut} \\
&= \sum_{t\in D_{pq}} \frac{1}{\eta^t+1} \left(\frac{1}{2} - \frac{i\sqrt{p}}{2}\right) + \sum_{t\in\mathbb{Z}_{pq}^*\backslash D_{pq}} \frac{1}{\eta^t+1} \left(\frac{1}{2} + \frac{i\sqrt{p}}{2}\right) \\
&= \sum_{t\in D_{pq}} \frac{1}{\eta^t+1} \left(\frac{1}{2} - \frac{i\sqrt{p}}{2}\right) + \left[\frac{(p-1)(q-1)}{2} - \sum_{t\in D_{pq}} \frac{1}{\eta^t+1}\right] \left(\frac{1}{2} + \frac{i\sqrt{p}}{2}\right) \\
&= \frac{(p-1)(q-1)}{4} + \frac{(p-1)(q-1)\sqrt{p}\,i}{4} - i\sqrt{p} \sum_{t\in D_{pq}} \frac{1}{\eta^t+1} \\
&= \frac{(p-1)(q-1)}{4} + \frac{(p-1)(q-1)\sqrt{p}\,i}{4} - i\sqrt{p} \sum_{v\in D_p} \sum_{y\in\mathbb{Z}_q^*} \frac{1}{\lambda^y\mu^v+1}.
\end{aligned}$$

By Lemma 2.1, we get

$$\begin{aligned}
&\sum_{v\in D_p}\sum_{y\in\mathbb{Z}_q^*}\frac{1}{\lambda^y\mu^v+1}\\
&=\sum_{v\in D_p}\sum_{y\in\mathbb{Z}_q^*}\frac{\sum_{k=0}^{q-1}(-1)^k\lambda^{ky}\mu^{kv}}{1+\mu^{qv}}\\
&=\sum_{v\in D_p}\frac{1}{1+\mu^{qv}}\sum_{k=0}^{q-1}(-1)^k\mu^{kv}\sum_{y\in\mathbb{Z}_q^*}\lambda^{ky}\\
&=\sum_{v\in D_p}\frac{1}{1+\mu^{qv}}\left(q-\sum_{k=0}^{q-1}(-1)^k\mu^{kv}\right)\\
&=q\sum_{v\in D_p}\frac{1}{1+\mu^{qv}}-\sum_{v\in D_p}\frac{1}{1+\mu^{v}}\\
&=\begin{cases}(q-1)\left[\frac{p-1}{4}+\frac{p-1}{4}\sqrt{p}\,i-\sqrt{p}\,i\Delta\right], & \text{if } \left(\frac{q}{p}\right)=1\\ (q-1)\frac{p-1}{4}-(q+1)\frac{p-1}{4}\sqrt{p}\,i+(q+1)\sqrt{p}\,i\Delta, & \text{if } \left(\frac{q}{p}\right)=-1\end{cases}
\end{aligned}$$

Combining all cases, we have

$$o_{1,1}=\begin{cases}(p-1)(q-1)/4, & \text{if } \left(\frac{q}{p}\right)=1;\\ (p-1)(q-1)/4-\frac{p-1}{2}+2\Delta, & \text{if } \left(\frac{q}{p}\right)=-1.\end{cases}$$

□

From Lemma 2.2, the case $n=p^r\times q^s$ follows readily.

Lemma 2.3 *Let $n=p^r\times q^s$, where r and s are positive integers, p and q are primes satisfying $p\equiv-1\pmod 8$ and $q\equiv\pm3\pmod 8$, respectively, and $\mathrm{ord}_n(2)=\varphi(n)/2$. Then the number of odd integers in $D_n=\{2^i\pmod n: i=0,1,\ldots,\mathrm{ord}_n(2)\}$, denoted by $o_{r,s}$, is*

$$o_{r,s}=\begin{cases}(p-1)(q-1)p^{r-1}q^{s-1}/4, & \text{if } \left(\frac{q}{p}\right)=1;\\ (p-1)(q-1)p^{r-1}q^{s-1}/4-(p-1)/2+2\Delta, & \text{if } \left(\frac{q}{p}\right)=-1,\end{cases}$$

and the number of odd integers in $\mathbb{Z}_n^*\backslash D_n$,denoted by $o'_{r,s}$, is

$$o'_{r,s}=\begin{cases}(p-1)(q-1)p^{r-1}q^{s-1}/4, & \text{if } \left(\frac{q}{p}\right)=1;\\ (p-1)(q-1)p^{r-1}q^{s-1}/4+(p-1)/2-2\Delta, & \text{if } \left(\frac{q}{p}\right)=-1,\end{cases}$$

where Δ is the number of odds in D_p.

Proof. When $n=p^r\times q^s$, we denote set

$$I_{r,s}=\left\{x+pqk: x=2^j \bmod pq, 0<x<pq \text{ and } 0\le k<p^{r-1}\times q^{s-1}\right\}.$$

Since $2^{\frac{(p-1)(q-1)}{2}i+j} \equiv 2^j(1+pqt)^i \equiv 2^j+pqk \pmod{p^r q^s}$ for some integer $0 \le k < p^{r-1}\times q^{s-1}$, where $0 \le i < p^{r-1} \times q^{s-1}$ and $0 \le j \le \frac{(p-1)(q-1)}{2}$, it follows that $D_n \subseteq I_{r,s}$. And $D_q = I_{r,s}$ since $|D_q| = |I_{r,s}|$. And note a number $x + pqk \in I_{r,s}$ is odd if and only if x and k are of different parity. By Lemma 2.2, and

$$o_{r,s} = o_{1,1} \times \frac{p^{r-1} \times q^{s-1} + 1}{2} + \left(\frac{(p-1)(q-1)}{2} - o_{1,1}\right) \times \frac{p^{r-1} \times q^{s-1} - 1}{2},$$

the computation of the result is straightforward. □

Theorem 2.1 *For positive integers r and s, let $\mathcal{S}$ denote a binary half-ℓ-sequence for which the connection integer is given by $n = p^r \times q^s$, where $p \equiv -1 \pmod 8$ and $q \equiv \pm 3 \pmod 8$. Then*

$$A_{\mathcal{S}}^{A}(\tau) \in \left\{0, \pm p^{r-i}q^{s-j}(p-1-4\Delta), \pm\frac{1}{2}p^{r-i+t}q^{s-1}(q-1)(p-1-4\Delta)\right\},$$

where $1 \le i \le r$, $1 \le j \le s$, $0 \le t < i$, and Δ is the number of odds in D_p.

Proof. For connection integer $n = p^r q^s$. If n' is a factor of n, then n' belong to

$$\left\{p^i, q^j, p^i \times q^j\right\},$$

for $1 \le i \le r$, $1 \le j \le s$.

(i) When $q' = p^i \times q^j$ with $1 \le i \le r$ and $1 \le j \le s$, by (1) and Lemma 2.3 we have

$$A_{\mathcal{S}}^{A}(\tau) = p^{r-i}q^{s-j}\left(\mathrm{ord}_{p^iq^j}(2) - 2\lambda'\right) = \begin{cases} 0, & \text{if } \left(\frac{q}{p}\right) = 1 \\ \pm p^{r-i}q^{s-j}(p-1-4\Delta), & \text{if } \left(\frac{q}{p}\right) = -1 \end{cases}$$

(ii) When $q' = p^i$ with $1 \le i \le r$, by (1) and [2, Thm. 1] we have

$$A_{\mathcal{S}}^{A}(\tau) = p^{r-i}q^{s-1}(q-1)\left(\mathrm{ord}_{p^i}(2) - 2\lambda'\right) \in \left\{\pm\frac{1}{2}p^{r-i+t}q^{s-1}(q-1)(p-1-4\Delta)\right\},$$

where $0 \le t < i$.

(iii) When $q' = q^j$ with $1 \le j \le s$, by (1)we have

$$A_{\mathcal{S}}^{A}(\tau) = 0,$$

since $\mathrm{ord}_{q^j}(2) = q^{j-1}(q-1)$. This establishes the results. □

Using [12, Lem. 5] and Theorem 2.1, an upper bound for the arithmetic-correlation can be established.

Corollary 1 *For positive integers r and s, let $\mathcal{S}$ denote a binary half-ℓ-sequence for which the connection integer is given by $n = p^r \times q^s$, where $p \equiv -1 \pmod 8$ and $q \equiv \pm 3 \pmod 8$. Then*

$$\left|A_{\mathcal{S}}^{A}(\tau)\right| \le \frac{1}{2}p^{r-1}q^{s-1}(q-1)\left[p - 1 + 4\left(0.3 + \frac{1}{\pi}\ln\frac{p-1}{2}\right)\left(p^{\frac{1}{2}}+1\right) - \frac{(p-1)(p-2)}{p}\right],$$

when $\tau \neq 0$.

2.2 When $p \equiv 3 \pmod 8$ and $q \equiv -3 \pmod 8$

Using methods analogous to those in Lemma 2.3 and Theorem 2.1, we can derive two corollaries for the case where $p \equiv 3 \pmod 8$ and $q \equiv -3 \pmod 8$.

Corollary 2 *Let $n = p^r \times q^s$, where r and s are positive integers, p and q are primes satisfying $p \equiv 3 \pmod 8$ and $q \equiv -3 \pmod 8$, respectively, and $\mathrm{ord}_n(2) = \varphi(n)/2$. Then the number of odd integers in $D_n = \{2^i \pmod n : i = 0, 1, \ldots, \mathrm{ord}_n(2)\}$, denoted by $\varepsilon_{r,s}$, is*

$$\varepsilon_{r,s} = (p-1)(q-1)(p^{r-1}q^{s-1} - 1)/4 + \Lambda,$$

and the number of odd integers in $\mathbb{Z}_n^* \backslash D_n$, denoted by $\varepsilon'_{r,s}$, is

$$\varepsilon'_{r,s} = (p-1)(q-1)(p^{r-1}q^{s-1} + 1)/4 - \Lambda,$$

where Λ is the number of odds in $D_{pq} = \{2^i \pmod{pq} : i = 0, 1, \ldots, \mathrm{ord}_{pq}(2)\}$.

Corollary 3 *For positive integers r and s, let $\mathcal{S}$ denote a binary half-ℓ-sequence for which the connection integer is given by $n = p^r \times q^s$, where $p \equiv 3 \pmod 8$ and $q \equiv -3 \pmod 8$. Then*

$$A^A_{\mathcal{S}}(\tau) \in \left\{0, \pm p^{r-i}q^{s-j}\left(\frac{(p-1)(q-1)}{2} - 2\Lambda\right)\right\},$$

where $1 \leq i \leq r$, $1 \leq j \leq s$, and Λ is the number of odds in $D_{pq} = \{2^i \pmod{pq} : i = 0, 1, \ldots, \mathrm{ord}_{pq}(2)\}$.

Some examples are listed in Table 1 to confirm the main results.

Table 1: $A^A_{\mathcal{S}}(\tau)$ of binary half-ℓ-sequence $\mathcal{S}$ with connection integer $n = p^r \times q^s$.

$n = p^r \times q^s$	period T	$\lvert A^A_{\mathcal{S}}(\tau)\rvert \in$
$69 = 23 \times 3$	22	$\{0, 6\}$
$207 = 23 \times 3^2$	66	$\{0, 18\}$
$115 = 23 \times 5$	44	$\{0, 6, 12\}$
$141 = 47 \times 3$	46	$\{0, 10\}$
$423 = 47 \times 3^2$	138	$\{0, 30\}$
$55 = 11 \times 5$	20	$\{0, 4\}$
$507 = 3 \times 13^2$	156	$\{0, 4\}$

3 Conclusions

In this paper, we studied arithmetic autocorrelations of binary half-ℓ-sequences with connection integers $n = p^r \times q^s$. If $p \equiv -1 \pmod 8$ and $q \equiv \pm 3 \pmod 8$, the arithmetic autocorrelations of the sequence depend on the number of odd integers in the cyclic subgroup generated by 2 modulo p. By appropriately selecting primes p that satisfy the required conditions, the arithmetic correlation value can be minimized. In fact, our results are also applicable to the arithmetic correlation of sequences with the same period.

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